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ORIGINAL RESEARCH

Polyacrylic acid/carboxymethyl cellulose/activated carbon composite hydrogel for removal of heavy metal ion and cationic dye

Xiaohong Wang · Xiangbo Fan · Honghao Xie · Xin Li · Chen Hao

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Abstract For the purpose of purifying polluted water, polyacrylic acid grafted carboxymethyl cellulose and activated carbon composite material (PAA/CMC/AC) was used to efficiently and selectively capture heavy metal ions and dyes in polluted water. The maximum adsorption capacity of Cu^{2+} and methylene blue (MB) in the independent system reached 169 mg g^{-1} and 119.9 mg g^{-1}, respectively. It is worth noting that in the case of coexistence of Cu^{2+} and MB, the adsorption of MB is hardly affected, while adsorption capacity of the adsorbent for Cu^{2+} increases with the increase of MB concentration in the presence of dyes, from 169 to 193.5 mg g^{-1}. For this, the adsorption isotherms and kinetics were fitted well by the Freundlich model and pseudo-second-order kinetic model. XPS results confirmed that surface complexation and electrostatic attraction are the main mechanisms of adsorption reaction. Finally, the regeneration performance of the adsorbent was explored.

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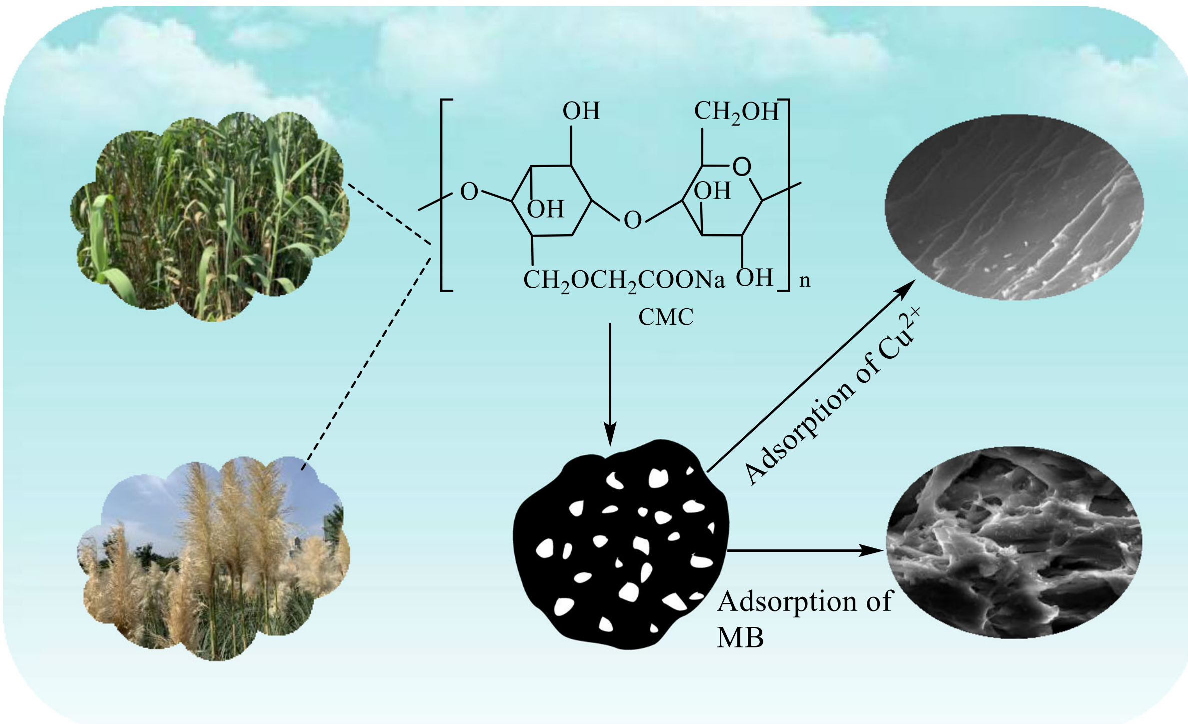
Graphic abstract Adsorption properties of Polyacrylic acid/carboxymethyl cellulose/activated carbon composite hydrogel for heavy metal ion and cationic dye



Keywords Adsorption · Heavy metal ions · Dyes · Selectively capture

Introduction

The shortage of water resources has always been a life-related problem facing human beings, and an important cause of water shortages is the problem of water pollution (Awual et al. 2019; Fu and Huang 2018). Therefore, calling for people to reuse water resources is being implemented. Most of the sources of water pollution come from the discharge of wastewater containing heavy metal ions and organic dyes in industrial production. The accumulation of heavy metal ions in the food chain will eventually cause them to enter the human body and endanger human health, while the wastewater containing dyes is discharged into the environment, which would not only cause visual pollution but also affect the survival of animals and plants in the water (Lin et al. 2021; Qin et al. 2019; Yu et al. 2018; Zhang et al. 2018). However, most of the previous studies have focused on the removal of single-class heavy metal ions and dyes, which limits their practical utility because it is impossible to have only one type of pollutant due to the complex composition of wastewater. It is worth noting that heavy metal ions can be selectively recovered from complex wastewater as a valuable resource, providing huge economic incentives for pollution control (Saeed et al. 2020; Zhang et al. 2020). In recent years, research on industrial wastewater treatment has focused on the synthesis of functional adsorbents because of their environmental, low cost and high efficiency. It is concluded that the development of high-performance selective adsorbents has become a development trend, which will be a huge challenge for the academic community (Wang et al. 2018a; Yang et al. 2019a).

Materials capable of adsorbing pollutant should have certain characteristics. First, it should be low cost and easy to manufacture. Secondly, it should contain abundant oxygen sites (hydroxyl, carboxyl, etc.) in its structure, and it has strong complexation ability for heavy metal. This chemical structure can provide either a chelation site or an ion exchange group. Furthermore, it should maintain a hydrophilic network structure to support the extended flow of water (Godiya et al. 2019; Kong et al. 2020).

As a hydrogel matrix with high water solubility, wide source and low cost, carboxymethyl cellulose (CMC) has various effective functional groups (Abouzeid et al. 2019; Bahadoran Baghbadorani et al. 2020). However, the initial CMC hydrogel has low adsorption capacity for heavy metal ions and dyes due to the lack of effective active sites, so it is necessary to introduce functional groups by chemical reaction or graft reaction and introduce side chains to improve adsorption efficiency (Lu et al. 2019; Zhou et al. 2020). In previous studies, it has been shown that CMC-based hydrogels have been widely used in the treatment of heavy metal ions in wastewater. Honglei Fan et al. have prepared the CMC-Fe_3O_4 nanoparticles by one-step method for highly efficient removal of heavy metal ions, they found that in situ nucleation of Fe_3O_4 nanoparticles on the CMC chain significantly improved the ability to handle heavy metal ions (Fan et al. 2019). Li Song et al. have preparared the CMC-PEI hydrogel to improve the removal rate of Cr(VI), they observed that the adsorption capacity is stronger under mild acidic conditions, and the efficient removal of Cr(VI) is hardly affected by coexisting substances. The hydrogel has great practical application potential (Song et al. 2019). Chirag B Godiya et al. have prepared a new bio-based hydrogel with dual functions in wastewater treatment and catalytic applications. The Cu(II) ions adsorbed in the hydrogel were reduced in situ to obtain a CMC/PAM hydrogel carrying copper NPs. This hydrogel successfully reduced 4-nitrophenol to 4-aminophenol (Godiya et al. 2019).

Powdered activated carbon (AC) is made from high-quality wood chips, coconut shell, and is processed through a series of production processes. In the past reports, AC has been widely used as an adsorbent for the removal of heavy metal ions. It is a very promising technology, but its application in wastewater treatment is greatly affected because of its difficult separation and poor selectivity. (Dong et al. 2018; Li et al. 2019). Therefore, functionalizing and modifying the original AC into an advanced composite with new structure and surface properties is the key to improving its environmental applicability (Yang et al. 2019b; Zhang et al. 2019). Uniform impregnation of CMC on the AC surface is an effective method for removing heavy metal ions to overcome this disadvantage. This method uses AC as the ideal support frame, which is mainly due to the complex internal porous structure formed during the activation process and its affinity for most of the compounds. CMC provides an active site for the chelation of heavy metal ions by including both a carboxyl group and a hydroxyl group in its structure (Hong et al. 2019; Jiang et al. 2020). The carboxyl group monomer acrylic acid is grafted on CMC and AC mixture to form polymer hydrogel by a free radical chain reaction, which is called PAA/CMC/AC.

As a toxic dye, MB is difficult to remove because it has a certain tolerance to acid and alkali (Wu et al. 2019). In this experiment, two typical toxic pollutants are used as research objects, which are dye MB and Cu^{2+}. Not only the adsorption capacity of PAA/CMC/AC in single system, but also the interaction of the two pollutants in the binary system were investigated.

Herein, a simple one-step radical polymerization method was used to synthesize a composite hydrogel based on acrylic monomers, and the composite gel consists of the crosslinked polyacrylic acid with carboxymethyl cellulose and AC, which have good ability to remove heavy metal ions and organic pollutants. By exploring the influence of various factors on the adsorption capacity, the adsorption mechanism was revealed from the perspective of adsorption dynamics and adsorption isotherms.

Experimental

Chemicals and materials

The reagents used in this experiment are of analytical grade, of which CMC, acrylic acid (AA), AC, potassium persulfate (KPS), absolute ethanol, NaOH, HCl, HNO_3, $Cu(NO_3)_2{\cdot}3H_2O$, $Pb(NO_3)_2$,

$Ni(NO_3)_2{\cdot}6H_2O$, $Co(NO_3)_2{\cdot}6H_2O$, $Cd(NO_3)_2{\cdot}4H_2O$ and $Cr(NO_3)_3{\cdot}9H_2O$ are purchased from Sinopharm Group Co., Ltd. Tianjin Kemeiou Chemical Reagent Co., Ltd. provides the N, N-methylenebisacrylamide (NMBA) needed for the experiment.

Synthesis of the PAA/CMC/AC composite adsorbent

Pretreatment of AC: First, 2 g of AC was washed with deionized water and dried in an oven at 120 °C for 12 h. Then, the dried AC was poured into 100 ml 25% nitric acid solution and refluxed in a water bath at 75 °C for 2 h. Finally, the oxidized AC was repeatedly washed to neutral and then dried in an oven at 120 °C for standby.

Preparation of PAA/CMC/AC adsorbent: First, 2.830 g KOH was poured into 5 mL deionized water to fully dissolve it. Then, under the condition of ice bath, 5 mL AA was slowly added to KOH solution while stirring. After the above solution was cooled to room temperature, 0.02 g of AC was added and stirred well for 30 min to make it mixed evenly. Finally, 0.06 g CMC was dissolved in 10 mL deionized water and mixed with the above solution, then 0.0289 g KPS and 0.0173 g NMBA were added to the above solution and treated in ultrasonic water bath for 1 h to obtain hydrogel. The sample was dried to a constant weight at 80 °C and pulverized for subsequent experiments. Scheme 1 shows the specific synthesis mechanism.

Experimental design and optimization by response surface methodology (RSM)

RSM is a statistical method to find the optimal process parameters and solve multivariable problems. The main goal of RSM is to find the optimal value of variables and maximize the response. In this study, RSM is used to define the effects of four main parameters on response (adsorption capacity) as a tool for modeling and optimization. These variables are CMC(A), KPS(B) and NMBA(C). The regression model equation is as follows:

$$Y = \alpha_0 + \sum_{i=1}^{n} \alpha_i x_i + \sum_{i=1}^{n} \alpha_{ii} x_{ii}^2 + \sum_{i=1}^{n-1} \sum_{j=2}^{n} \alpha_{ij} x_i x_j + \delta \quad (1)$$

where Y is the dependent variable; α_0 is a constant; α_i, α_{ii} and α_{ij} are the coefficients of linear term, quadratic term and interactive term respectively; x_i and x_j are independent variables; δ is the error. The experimental data of BBD (Box-Behnken design) were fitted by regression model equation. Analysis of variance (ANOVA) was used to evaluate the significance of the model.

Characterization of adsorbents

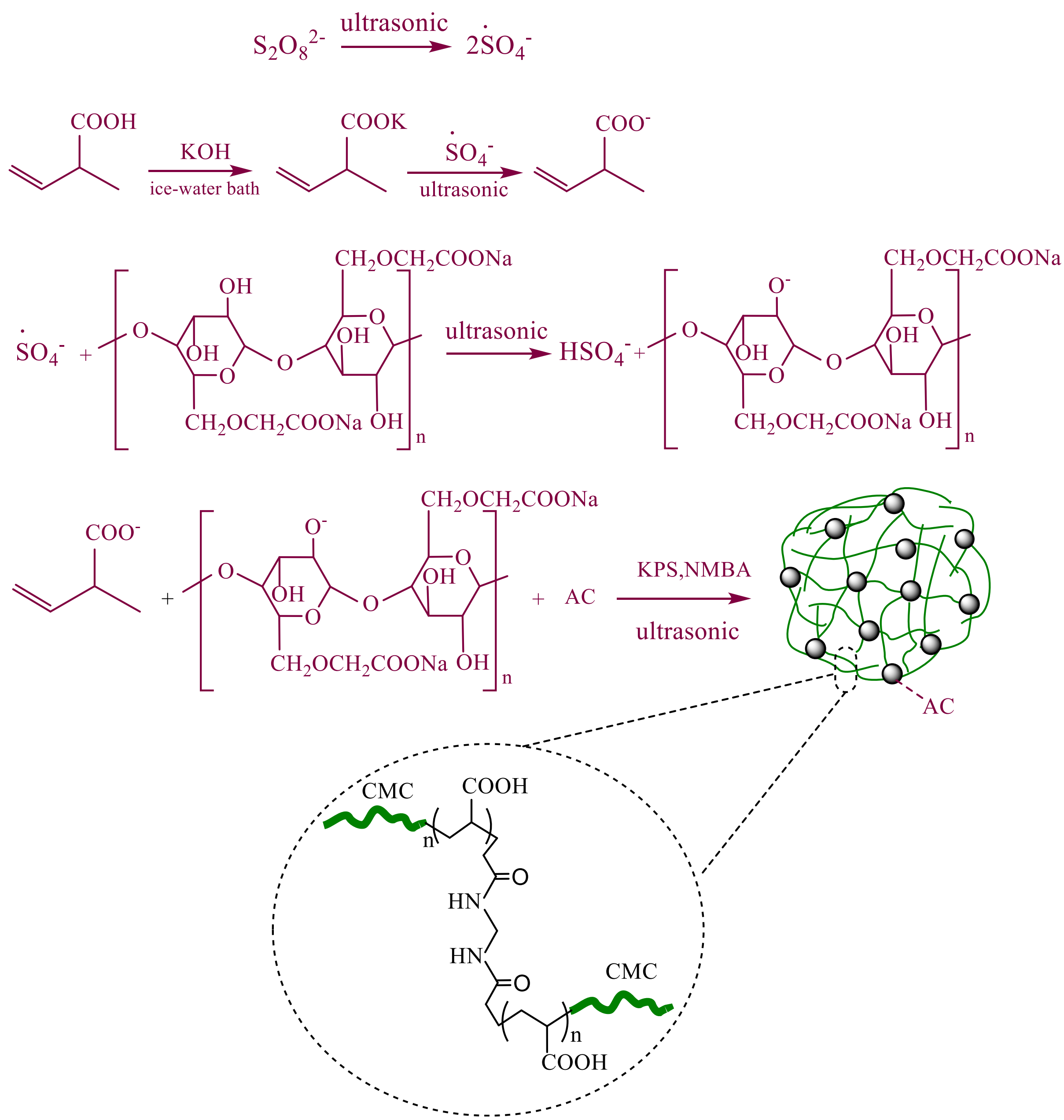
Fourier transform infrared (FTIR) spectra of resulting samples were recorded by using Nicolet Nexus 470 spectrometer (Nicolet, USA). Scanning electron microscope (SEM) and energy dispersive spectroscopy (EDS) (JSM-7800F, Jeol Ltd., Japan) were used to determine the surface morphology and elemental composition of the samples. Thermogravimetric (TGA) analysis was conducted via STA 449 C integrated thermal analyzer (Netzsch, Germany) under N_2 atmosphere. X-ray diffractometer (XRD) spectrum obtained from XRD-6100 (Japan). The Zeta potentials of the adsorbent at various pH were measured by nano ZS90 ζ potential analyzer (UK, Malvern). X-ray photoelectron spectroscopy (XPS, Thermo-Fisher-Scientific) was used to analyze the chemical state of the sample surface before and after adsorption.

Adsorption behavior in independent system

Adsorption experiments were carried out using the prepared adsorbent in mono-pollutant system. All experiments were performed in 50 mL beakers, and the effects on the absorption of pollutants were judged by evaluating adsorption kinetics, isotherm, solution pH, and the amount of adsorbent. The equilibrium adsorption capacity of the adsorbent for pollutants is determined by the following formula:

$$q_e = \frac{(C_0 - C_e)V}{m} \quad (2)$$

where C_0 is the initial bulk concentration, C_e is the residue concentration in solution (mg L^{-1}) at equilibrium, V represents the volume of the solution participating in the reaction (L), and m is the weight of the hydrogels (g).



Scheme 1 Reaction scheme for synthesis of the adsorbent

Effect of solution pH

The adsorption of two pollutants by PAA/CMC/AC was performed under different pH conditions. For the adsorption of Cu^{2+}, at 25 °C, 10 mg adsorbent was added to 50 mL working solution (50 mg L^{-1}) for 3 h, the initial solution pH ranged from 2 to 6. Similarly, 10 mg of adsorbent was added to 50 ml of dye solution (25 mg L^{-1}) at 25 °C for 24 h of MB adsorption and the initial solution pH ranged from 2 to 12.

Effect of sorbent dose

Experiments were performed with different amount of adsorbent, 10 mg to 60 mg of adsorbent was added to the Cu^{2+} solution (50 mg L^{-1}) and MB solution (25 mg L^{-1}) at 25 °C for 3 h and 24 h adsorption experiments, respectively.

Effect of time on adsorption capacity

The effect of time on the amount of adsorption was determined by measuring the concentration of the working solution at different times. The specific operation is as follows: 10 mg of PAA/CMC/AC was added into 50 mL of different pollutant solutions (the concentrations were 50 mg L^{-1} for Cu^{2+} and 25 mg L^{-1} for MB), the capture of Cu^{2+} lasted 210 min, and samples were taken at 10, 20, 30, 60, 90, 120, 150, 180, 210 min. The adsorption of MB was performed for 48 h, the solution was sampled and measured at 3, 6, 9, 12, 15, 18, 21, 24, 36 and 48 h, respectively.

Influence of pollutant concentration

Put 10 mg of adsorbent in 50 mL of pollutant solution, the initial concentrations ranged from 50 to 100 mg L^{-1} for Cu^{2+} and from 5 to 30 mg L^{-1} for MB. The rest of the conditions are the same as the effect of the previous study on the amount of adsorbent.

Adsorption behavior in binary system

The selective removal of heavy metal ions and organic dyes in binary system and their interaction in complex wastewater were studied. In the case of MB (15 and 25 mg L^{-1}) coexistence, Cu^{2+} with an initial concentration of 50–100 mg L^{-1} was adsorbed in the MB-Cu^{2+} binary system. Similarly, the adsorption amount of MB (5–25 mg L^{-1}) was measured in the presence of Cu^{2+} (15 and 25 mg L^{-1}) in Cu^{2+}-MB binary system. After that, the interaction between MB and heavy metals in the binary system was determined by comparing the resulting adsorption amounts of each pollutant with that obtained in mono-pollutant system under the same conditions.

Reusability studies

0.1 mol L^{-1} HCl was used as the desorption agent to carry out the desorption experiment on the pollutant loaded adsorbent. Then, wash the desorbed adsorbent repeatedly with deionized water and dry it for the next cycle adsorption experiment. The formula for calculating the efficiency of circulating adsorption is as follows:

$$\phi = \frac{C_0 - C_e}{C_0} \times 100\% \quad (3)$$

Here, ϕ refers to the cycle adsorption efficiency.

Analytical methods

According to the calibration curve (Fig. S1), the equilibrium concentration of the MB solution was measured at 664 nm with a UV–visible spectrometer (UV-3600Plus, Japan), and atomic absorption spectrophotometry (TAS-986, China) was used to determine the remaining content of Cu^{2+} in the solution. Each experiment was performed at least three times, and the averages were reported.

Results and discussion

Experimental design results of RSM

RSM is a method dedicated to the optimization of experimental conditions, which is suitable for solving related problems of nonlinear data processing. Table S1 shows the experimental schemes and corresponding response values (adsorption capacity for 100 mg L^{-1} of Cu^{2+}) designed by this method. Afterwards, ANOVA result for response surface quadratic model is shown in Table S2. Given that the F value of the model is 6.59, it indicates that the model is significant. Due to the influence of noise, there is only a 1.06% chance of such a large “F-value model”. Another condition that indicates the model is significant is that the values of “Prob > F” is less than 0.0500, so C, AC, and A^2 satisfy this condition. A negative “Pred R-Squared” implies that the overall mean is a better predictor of the response than the current model. Finally, “Adeq Precision” measures the signal-to-noise ratio and the value of F greater than 4 is desirable. It can be seen that the value is 10.698, indicating that the signal is adequate and the model can be used to navigate the design space. In addition, the influence relationship of various factors on the response value is shown in Fig. S2 in the support information. The final equation in terms of coded factors and actual factors are shown below:

$$q_e = 191.54 + 1.13A - 2.42B + 10.60C - 3.62AB + 12.94AC + 6.81BC + 10.83A^2 + 2.84B^2 - 3.00C^2 \quad (4)$$

$$q_e = 678.9 - 3707.53A - 25007.54B - 14821.34C - 59603.28AB + 217845AC + 752332BC + 27083A^2 + 305106B^2 + 340016C^2 \quad (5)$$

Characterization of adsorbents

Morphology of PAA/CMC/AC hydrogel

SEM images were collected to determine the morphology of the prepared PAA/CMC/AC. From Fig. 1, it can be clearly seen that there are obvious differences in the surface morphology of the sample before and after adsorption. Figure 1a shows the PAA/CMC/AC image before adsorption, a large amount of pores were observed. In contrast, the surface of the sample becomes rough, wrinkled and attached with particles after adsorption as shown in Fig. 1b. Compared with the adsorption of heavy metal ions, the difference is that the surface of the sample shows a completely different morphology after adsorption of MB, it is composed of many thick irregular crisscrossing strips with uneven surfaces, which is displayed in Fig. 1d. Moreover, it can be seen from Fig. 1c that many fine particles are attached to the adsorbent. Synchronously, EDS spectra (Fig. S4) records the element composition of the PAA/CMC/AC, which is consistent with our expected results.

FTIR analysis

Infrared analysis is an important technology for identifying functional groups on the surface of adsorbents, which can help us further explore the changes of functional groups before and after adsorption. As shown in Fig. 2a, FTIR results show that PAA/CMC/AC composite hydrogel have characteristic peaks of both CMC and AC functional groups. The FTIR spectrum of the CMC shows characteristic peaks at 1328 cm^{-1} corresponding to the stretching vibrations of C–O (Fan et al. 2019). Moreover, carboxylate compounds have obvious O–C=O asymmetric and symmetric stretching vibration at 1620–1540 and 1420–1390 cm^{-1} (Zhang et al. 2019), this type of spectral bands were observed in the FTIR spectrum of PAA/CMC/AC, which means that the carboxyl group

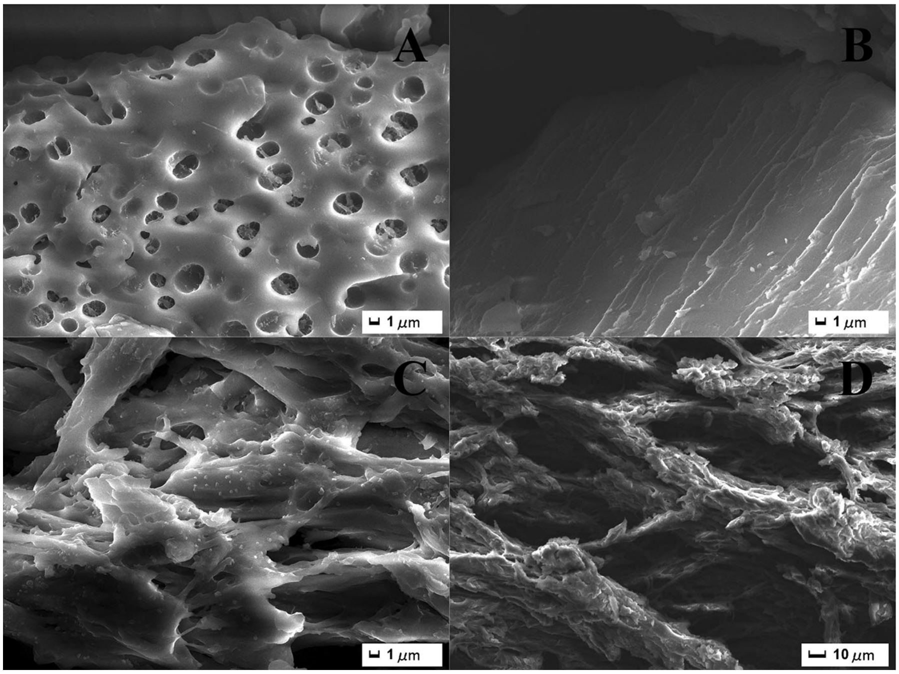


Fig. 1 SEM images of PAA/CMC/AC (**a**), PAA/CMC/AC-Cu^{2+} (**b**), PAA/CMC/AC-MB (**c**–**d**)

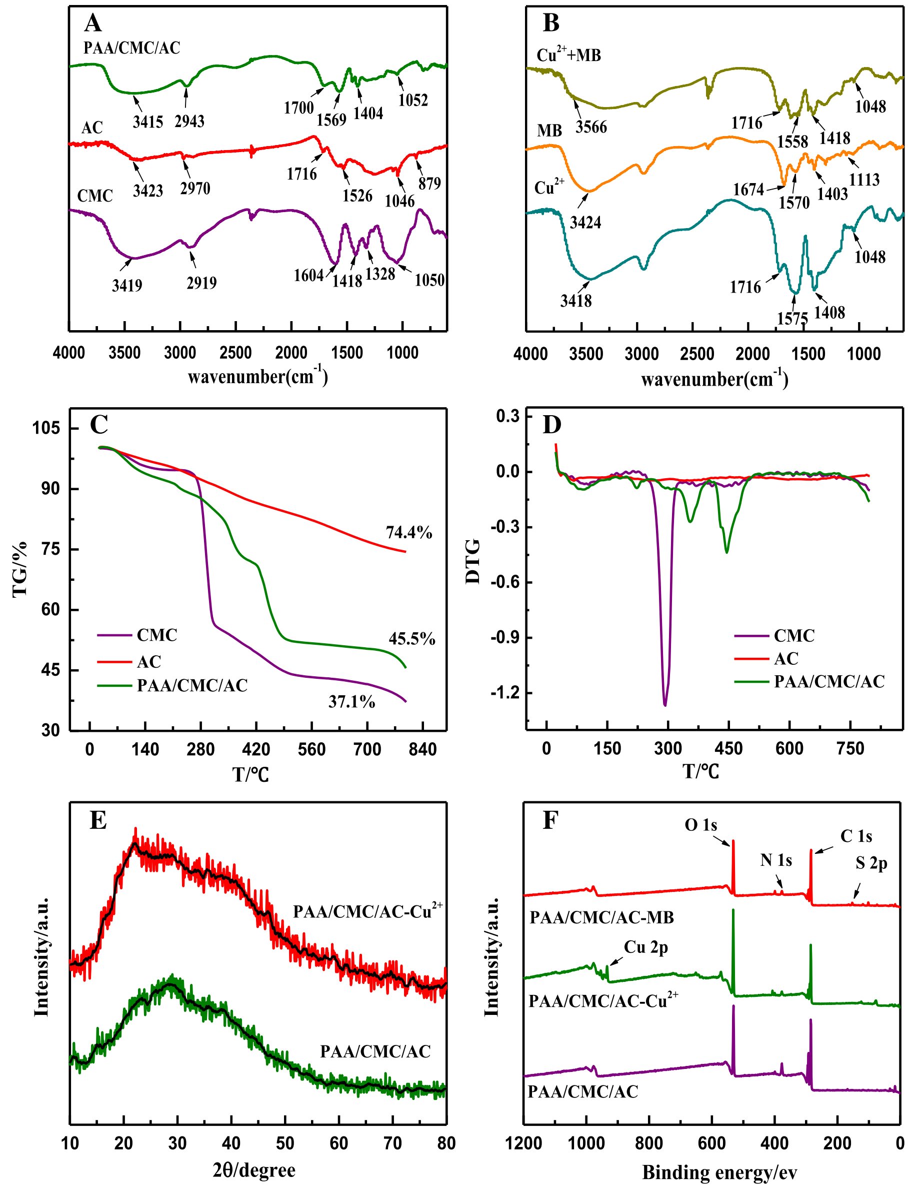


Fig. 2 FTIR spectra of CMC, AC and PAA/CMC/AC (**a**) and the adsorption products: adsorption of Cu^{2+}, adsorption of MB, co-adsorption of Cu^{2+} and MB (**b**); TG (**c**) and DTG (**d**) curves of CMC, AC and PAA/CMC/AC; XRD patterns of the PAA/CMC/AC and PAA/CMC/AC-Cu^{2+} (**e**); XPS spectra of the PAA/CMC/AC before and after adsorption (**f**)

is widely grafted on the composite, another characteristic absorption peak at ~ 1700 cm^{-1} may be caused by the stretching vibration of C=O on a typical carboxylic acid group. There is an absorption peak at 1569 cm^{-1} that can be attributed to the stretching vibration of N–H of the amide. Synchronously, the absorption peaks of hydroxyl group and carboxyl group also appeared after the activated carbon was oxidized. The band of PAA/CMC/AC at 1100–1000 cm^{-1} belongs to the tensile vibration of C–OH, obviously decreased in intensity compared with that of CMC and AC, which proves that the hydroxyl group were involved in the polymerization, besides, C–H stretching vibration in the FTIR spectrum of PAA/CMC/AC was also obviously shifted compared with that of CMC, from 2919 to 2943 cm^{-1}, these results also suggest the possible formation of intermolecular and intramolecular hydrogen bonds.

As shown in Fig. 2b, it can be clearly noticed that the vibration peak at 1404 cm^{-1} assigned to the carboxyl group shifts to 1408 cm^{-1} after adsorption of Cu^{2+} in the FTIR spectrum, which may be due to the coordination of the carboxyl group with Cu^{2+}. A new vibration signal appeared at 1113 cm^{-1} which can be attributed to the C–S stretching vibration peak of MB. Meanwhile, the apparent shift of the peak at 3415 cm^{-1} (the stretching vibration of O–H) and 1569 cm^{-1} (the bending vibration of N–H) further indicates that active functional groups such as –OH, –NH_2 play a key role in adsorption.

TGA analysis

The thermal stability of hydrogel will affect its application in practical environment. Therefore, the thermal stability of hydrogel adsorbent was studied by thermogravimetric analysis. As shown in Fig. 2c–d, the results showed that the total weight loss rates of CMC, AC and PAA/CMC/AC were 62.9%, 25.6% and 54.5% in the studied temperature range, respectively. Based on the weight percentage change, the thermal decomposition behavior of CMC can be divided into two regions, apart from water loss below 250 °C, above 250 °C was mainly the decomposition of CMC backbone and the removal of oxygen-containing functional groups. For AC, the weight loss in the range of room temperature to 800 °C comes entirely from the evaporation of physically adsorbed or structured water in AC. TGA analysis shows that PAA/CMC/AC has three decomposition stages, the incipient declines in the range of 25–250 °C with a mass loss of 11.29% can be ascribed to the evaporation of adsorbed water and moisture, the second step of degradation occurred in the range of 250–410 °C with a mass loss of 16.77%, which may be due to the chain break of the graft polymer and the pyrolysis of the polymer, while the significant weight loss from 410 to 800 °C resulted from the decomposition of the polymer backbone, when the temperature reached 800 °C, all cross-linked chains collapse and the polymer is expected to break completely.

XRD analysis

Figure 2e compares the XRD spectra of samples before and after adsorption of Cu^{2+}. It can be observed that the diffraction angle at 22° belongs to amorphous carbon in PAA/CMC/AC and there is no additional characteristic peak, which confirms the amorphous nature of the adsorbent. Also, the weak signal peaks at around 43° and 73° were found in the spectrum after adsorption, which was caused by the loading of Cu^{2+}.

XPS analysis

XPS can be used to analyze the surface chemical properties of substances, including element composition, element chemical state and electronic state. Figure 2f shows the XPS spectrum of the adsorbent before and after adsorption, it can be roughly observed that the binding energy at about 531.35 eV, 284.9 eV, 399.53 eV corresponds to O 1 s, C 1 s, N 1 s, respectively. After interaction with Cu^{2+}, the new binding energy at 933.82 eV is attributed to Cu 2p, which indicates that Cu^{2+} are successfully adsorbed by PAA/CMC/AC. For uptake of MB, due to the presence of S element in the MB molecule, a new binding energy was found to be allocated to S 2p around 164.34 eV, confirming that MB has been successfully adsorbed, which is consistent with the FTIR spectrum.

Adsorption of Cu^{2+} and MB in an independent system

The factors influencing removal

(1) pH

As an important parameter that affects the adsorption capacity, pH can affect both the morphology of the pollutants in solution and the surface charge of the adsorbent (Jiang et al. 2019b). The removal capacity of MB and Cu^{2+} are shown in Fig. 3a, in view of the formation of hydroxide precipitation under alkaline conditions, the pH was selected in the range of 2–6, apparently, it is not conducive to the capture of Cu^{2+} at pH < 3, attributing to strong electrostatic repulsions between high concentrations of H^+ and metal cations in the solution, the increase in pH is accompanied by a significant enhancement of the adsorption capacity can be attributed to deprotonation of the active groups such as carboxyl and hydroxyl groups on the surface of the adsorbent, which leads to more electrostatic interaction between the active site on PAA/CMC/AC and the metal ions (Huang et al. 2017). For capture of

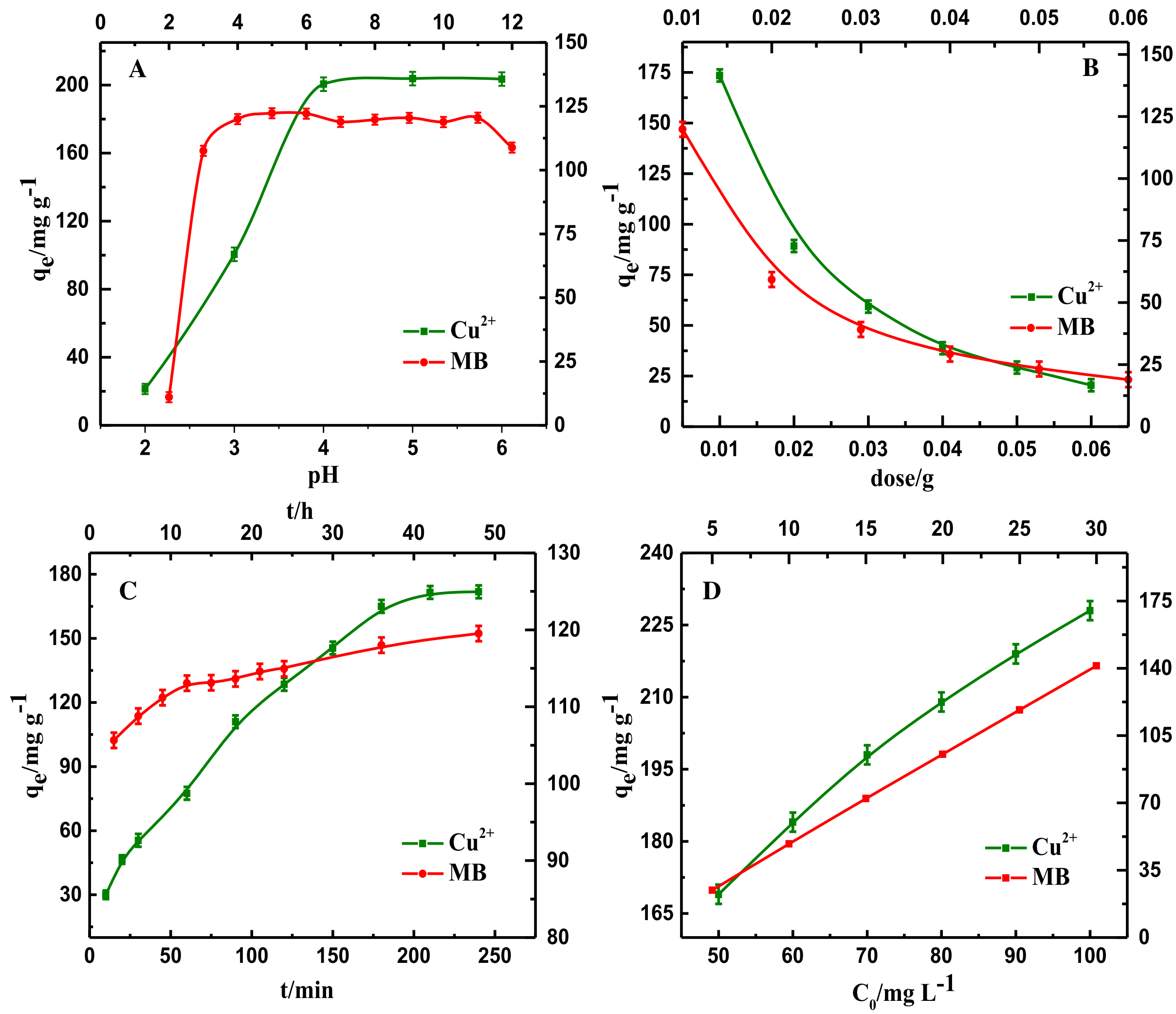


Fig. 3 The influence of four factors on the adsorption of Cu^{2+} and MB by PAA/CMC/AC: pH (**a**), the amount of adsorbent (**b**), contact time (**c**), concentration of pollutants (**d**)

MB, the amount of MB adsorbed by the adsorbent is basically stable except that the adsorption capacity is relatively weak at pH = 2. As shown in Fig. S5, the zeta potential of the adsorbent was measured under varying solution pH. Since heavy metal exist in the form of divalent cations, electrostatic repulsion inhibits the adsorption of metal ions and alkaline dye MB at low pH, as the zeta potential changes from positive to negative, the adsorbent with negative charge on the surface is easy to interact with Cu^{2+} and MB, resulting in an increase in the amount of adsorption (Qi et al. 2019; Zhou et al. 2018).

(2) Amount of adsorbent

In general, the amount of adsorbent directly affects the number of adsorption sites. The effective information that can be obtained from Fig. 3b is that as the mass of the adsorbent increases from 10 to 60 mg, the amount of pollutants adsorbed continues to decrease, obviously, the adsorption sites were not fully utilized in the case of high amount of adsorbent used, which may be caused by the agglomeration phenomenon that occurs when most adsorbents are in solution (Wang et al. 2020). In short, the use of excessive adsorbent in the actual sewage treatment cannot play its role well.

(3) Contact time

Similarly, as one of the major factors affecting adsorption capacity, the amount of pollutants adsorbed during the contact time determines the removal rate. From Fig. 3c, it can be intuitively observed that the adsorbent shows the same growth trend for the two pollutants, both of which are from rapid adsorption in the initial stage to the final equilibrium state. It can be explained that the initial rapid adsorption is due to the rich functional groups on the surface of the adsorbent providing a large amount of adsorption sites, not only that, PAA/CMC/AC shows low zeta potential with more negative charge on its surface at pH = 6.0, which has stronger electrostatic attraction towards two pollutants (Jiang et al. 2019a; Pu et al. 2018). Later, the adsorption gradually tended to equilibrium due to the exhaustion of the adsorption site.

(4) Concentration of pollutants

The change of the adsorption capacity with the initial concentration is shown in Fig. 3d, with the increase of concentration, the adsorption capacity also increased gradually. Perhaps some kind of interaction between the pollutants and the easily accessible sites on the surface of the adsorbent is the main reason for the difference in adsorption capacity. Additionally, large driving force for mass transfer made it easier for pollutants to bind to adsorption sites at high concentration (Wang et al. 2018b; Zare et al. 2018). The color change diagram of the solution before and after adsorption is placed in support information (Fig. S6).

Study on adsorption dynamics and adsorption isotherms

To further investigate the effect of time on adsorption capacity, the pseudo-first-order, pseudo-second-order kinetic and intraparticle diffusion models were used to analyze the adsorption data in detail, the three kinetic equations are shown below:

Pseudo-first order model:

$$\ln(q_e - q_t) = \ln q_e - k_1 t \tag{6}$$

Pseudo-second order model:

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \tag{7}$$

Intraparticle diffusion model:

$$q_t = k_p t^{0.5} + C \tag{8}$$

In these three equations, q_e and q_t correspond to the adsorption amounts at equilibrium time and at time t, respectively; k_1, k_2, k_p represent the adsorption rate constants of the pseudo-first-order, pseudo-second-order and intraparticle diffusion models, C is also a constant related to the intraparticle diffusion model.

The fitting kinetic diagrams of PAA/CMC/AC for Cu^{2+} and MB made by these three equations are shown in Fig. 4a–c, meanwhile, the relevant kinetic parameters can be learned from Table 1. Comparing the correlation coefficient R^2 obtained from the simulation of the three kinetic model equations, it is found that the value of the correlation coefficient R^2 obtained by the pseudo-second-order dynamic model is closer to 1. Thus, it can be said that the pseudo second-order kinetic model can better describe the adsorption behavior, and it also shows that chemical adsorption is the rate determination step of the adsorption process.

The information provided by the adsorption isotherm is the effect of the solution concentration on the adsorption capacity and the interaction between the adsorbed pollutants and the adsorbent. For this, four isotherm models are listed, the following are the equations of the four isotherm models.

Langmuir:

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e} \tag{9}$$

Freundlich:

$$q_e = K_F C_e^{1/n} \tag{10}$$

Temkin:

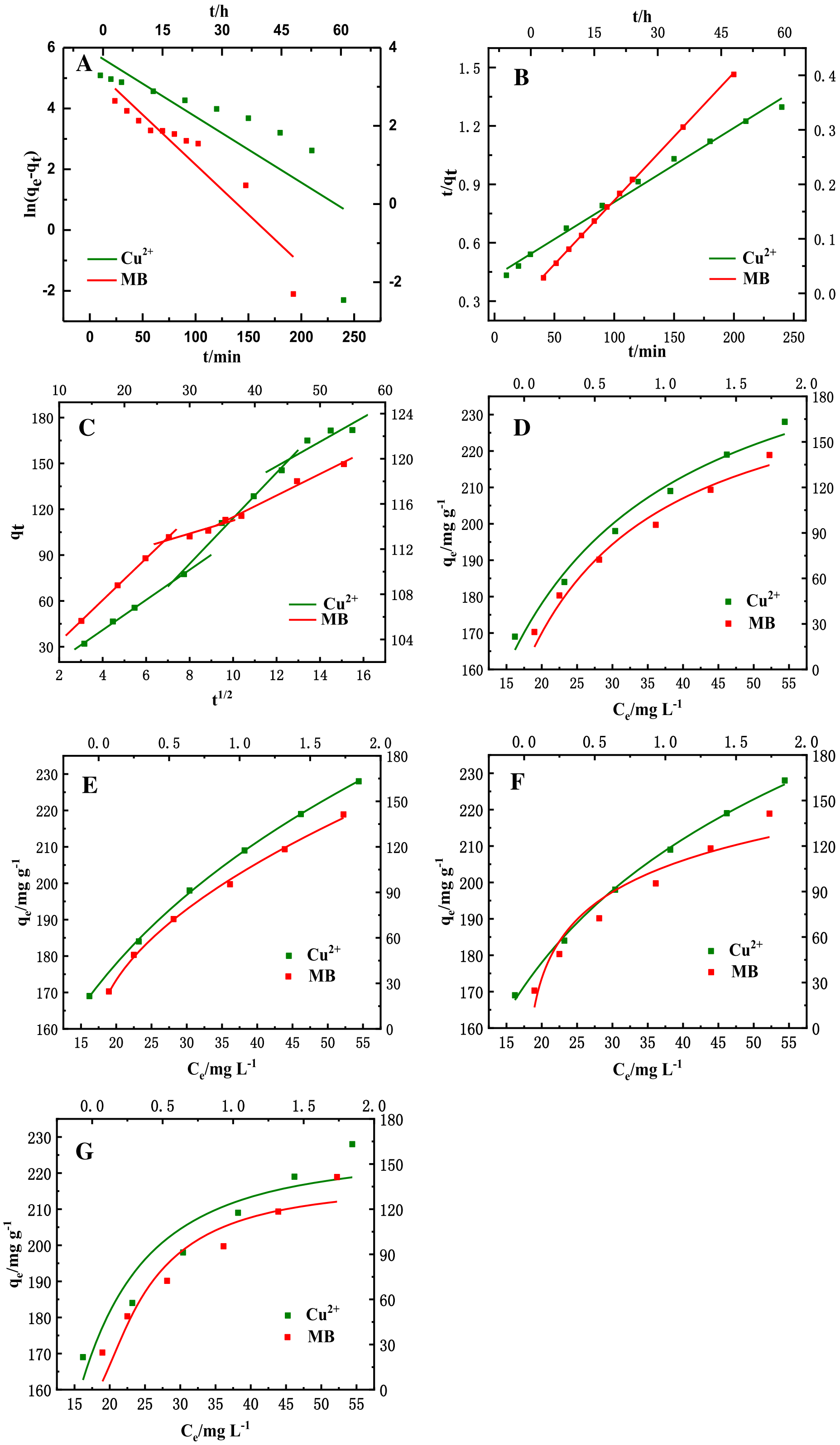
$$q_e = B \ln A + B \ln C_e \tag{11}$$

Dubinin-Radushkevich:

$$q_e = q_{m,D-R} e^{-\beta \varepsilon^2} \tag{12}$$

$$\varepsilon = RT \ln\left(\frac{C_e + 1}{C_e}\right) \tag{13}$$

In these four equations, C_e (mg L^{-1}) represents the concentration of adsorbed pollutants when adsorption reaches equilibrium, similarly, Q_{max} (mg g^{-1}) and q_e (mg g^{-1}) correspond to the maximum adsorption capacity and equilibrium adsorption capacity respectively. Besides, K_L and K_F are the Langmuir constant and Freundlich constant related to the adsorption


A
t/h
t/min
ln(qe-qt)
Cu2+
MB
B
t/h
t/min
t/qt
Cu2+
MB
C
t1/2
qt
Cu2+
MB
D
Ce/mg L-1
qe/mg g-1
Cu2+
MB
E
Ce/mg L-1
qe/mg g-1
Cu2+
MB
F
Ce/mg L-1
qe/mg g-1
Cu2+
MB
G
Ce/mg L-1
qe/mg g-1
Cu2+
MB

◀**Fig. 4** Adsorption kinetics of Cu^{2+} and MB by PAA/CMC/AC: pseudo-first-order kinetics (**a**) and pseudo-second-order kinetics (**b**), intragranular diffusion (**c**); adsorption isotherms: Langmuir (**d**), Freundlich (**e**), Temkin (**f**) and Dubinin-Radushkevich (**g**)

Table 1 Kinetic constants of the adsorbents for Cu^{2+} and MB

Type of pollutant	Cu^{2+}	MB
Pseudo-first-order model		
q_e (mg g^{-1})	361.4	25.57
k_1 (min^{-1})	0.022	0.095
R^2	0.623	0.861
Pseudo-second-order model		
q_e (mg g^{-1})	262.5	120.6
k_2 (g mg^{-1} min^{-1})	$3.339*10^{-5}$	0.010
R^2	0.992	0.999
Intraparticle diffusion		
Linear fitting		
k_p (mg g^{-1} $min^{-1/2}$)	12.37	0.293
C	− 9.447	104.2
R^2	0.987	0.850
Nonlinear fitting		
k_{p1} (mg g^{-1} $min^{-1/2}$)	1.560	0.564
C_1	9.849	98.07
R_1^2	0.998	0.999
k_{p2} (mg g^{-1} $min^{-1/2}$)	− 34.91	0.173
C_2	14.90	108.9
R_2^2	0.979	0.751
k_{p3} (mg g^{-1} $min^{-1/2}$)	8.031	0.287
C_3	51.73	104.3
R_3^2	0.735	0.979

capacity. A (L g^{-1}) and B are also constants about Temkin model. β (mol^2 kJ^{-2}) can be understand as a constant related to the energy of adsorption, ε is the potential of Polanyi, $q_{m,D-R}$(mg g^{-1}) represents the theoretical saturation capacity.

According to the four adsorption isotherm models, the fitting adsorption isotherms of PAA/CMC/AC toward Cu^{2+} and MB are illustrated in Fig. 4d–g, and the calculated related parameters can be seen from Table 2. For the adsorption process of these two pollutants on the adsorbent, the R^2 value of Freundlich isotherm model is higher than other isotherm models, which means that Freundlich model can better describe the adsorption process. That is, it shows that the adsorption of Cu^{2+} and MB by PAA/CMC/AC is a multi-molecular layer adsorption on heterogeneous surface. Meanwhile, the adsorption capacity of PAA/CMC/AC adsorbents for Cu^{2+} and MB is better than most of the adsorbents reported in the literature (shown in Table 3). This is because the combination of AC, PAA, and CMC greatly improves its utilization efficiency. Composite adsorbents have higher adsorption performance than single adsorbents.

Table 2 Isotherm parameters of the adsorbents for Cu^{2+} and MB

Type of pollutant	Cu^{2+}	MB
Langmuir		
Q_{max} (mg g^{-1})	264.9	209.1
K_L (L mg^{-1})	0.103	1.042
R^2	0.978	0.968
Freundlich		
K_F (L g^{-1})	84.6	102.3
$1/n$	0.248	0.556
R^2	0.999	0.997
Temkin		
A (mg L^{-1})	1.903	20.12
B	48.92	35.43
R^2	0.997	0.915
D-R		
$q_{m,D-R}$ (mg g^{-1})	225.5	137.0
β(mol^2 KJ^{-2}) $\times$ 10^{-3}	14.819	0.072
R^2	0.871	0.875

Adsorption mechanism

To reveal the interaction between the adsorbent and pollutants, XPS was used to analyze the chemical properties of the sample surface before and after adsorption. It can be seen from Fig. 5a that the C 1 s spectrum belonging to PAA/CMC/AC can be decomposed into three independent peaks with binding energy of 284.42 eV, 285.62 eV and 287.92 eV, which are assigned to the C atom in the form of C–C, C–O–C and O–C=O (Deng et al. 2017; Huang et al. 2019). Similarly, Fig. 5b shows that the O 1 s spectrum can also be divided into three separate peaks, corresponding to O–H (530.42 eV), C–O

Table 3 Adsorption capacity of Cu^{2+} and MB by different adsorbents

Adsorbents	Target pollutant	q_e (mg/g)	Ref.
CMC-Alg/GO	MB	78.5	(Allouss et al. 2019)
HA_x/CMC-yA	MB	76.92	(Chen et al. 2019a, b)
CMC/clay composite films	Cu^{2+}	54.6	(Gad et al. 2021)
GO-HBP-NH_2-CMC	Cu^{2+}	137.48	(Kong et al. 2020)
CSECM	Cu^{2+}	142.95	(Manzoor et al. 2019)
CMC coated Fe_3O_4@SiO_2 MNPs	MB	22.7	(Zirak et al. 2017)
PAA/CMC/AC	Cu^{2+}/MB	269/119.9	This study

(531.48 eV), C=O (532.65 eV) (Pan et al. 2019; Wang et al. 2017). Apart from this, C–N (399.71 eV), –NH– (399.3 eV) appears in the N 1 spectrum of Fig. 5c (Song et al. 2019). Figure 5d–k shows the XPS spectrum after adsorption, After interaction with Cu^{2+}, it is found that the binding energy of some groups has shifted, which is mainly manifested by the significant increase of the binding energy of C=O (532.91 eV) and O–H (531.91 eV) in the O 1 s spectrum, meantime, the binding energy of O–C=O (288.32 eV) belonging to C 1 s has also increased (Xia et al. 2020; Zhan et al. 2019). This may be due to the fact that Cu^{2+} form complexes with these groups, resulting in a reduction in the density of electron clouds around adjacent carbon atoms. Furthermore, a new binding energy appears at 407.03 eV which belongs to Cu–N in the N 1 s spectrum. From these results, it can be seen that the carboxyl group, hydroxyl group, and amino group are all involved in the adsorption of Cu^{2+}. More importantly, the Cu 2p spectrum was detected on the sample surface after adsorption, further illustrating the successful adsorption of Cu^{2+}. It can be clearly seen that the spectrum of Cu 2p can be divided into two peaks, corresponding to Cu–O at 932.7 eV and Cu–N at 934.36 eV, respectively (Liu et al. 2017; Xiao et al. 2019). For the adsorption of MB, the main performance is that the O–H (530.74 eV), O–C=O (288.22 eV) peak moves to a higher binding energy, which indicates that the cationic dye interacts with these functional groups. At the same time, the S 2p spectrum appears on the adsorbent, based on this information, it means that the MB was successfully adsorbed (Eltaweil et al. 2020; Ma et al. 2018). Figure 6 shows the mechanism of PAA/CMC/AC adsorbent capturing $Cu^{2+}$and MB.

Selective adsorption of the Cu^{2+} and MB

It is often necessary to remove many different types of pollutants at the same time in the actual water treatment process. Therefore, this study used the prepared adsorbent to remove heavy metal ions and dyes simultaneously and explore the interaction of these two pollutants in complex systems. Based on this purpose, the adsorption experiments in binary system of Cu^{2+}-MB and MB-Cu^{2+} were carried out to compare with that in mono-pollutant system.

Obviously, it can be seen from Fig. 7a that with the increase of the concentration of Cu^{2+} in the Cu^{2+}-MB system, the amount of MB adsorbed by the adsorbent hardly changes, demonstrating that the presence of Cu^{2+} does not affect the adsorption of MB, because PAA/CMC/AC has a larger adsorption capacity for the lower concentration of MB in the mixed solution. In contrast, the removal of Cu^{2+} by PAA/CMC/AC is enhanced in the case of MB coexistence. From Fig. 7b, it can be found that when the concentration of MB in the mixed solution increases from 0 to 25 mg L^{-1}, the corresponding adsorption capacity for Cu^{2+} also increases. Compared with the single system, the adsorption capacity of the adsorbent for Cu^{2+} increased from 169 to 193.5 mg g^{-1}. The reason for this phenomenon may be that the adsorbent provides two different adsorption sites for Cu^{2+} uptake, one is that Cu^{2+} interact with the hydroxyl and carboxyl groups on PAA/CMC/AC to form a complex to achieve the purpose of removal, moreover, When MB was adsorbed onto the adsorbent, the uptake of Cu^{2+} would be facilitated by the interaction on new sites derived from the –NH_2 and –NH– groups of MB. thereby improving the ability to capture Cu^{2+} (Chen et al. 2019a, b; Ge et al. 2019). In addition, Fig. S7 in support information shows that the adsorbent also has good adsorption performance for other metal ions.

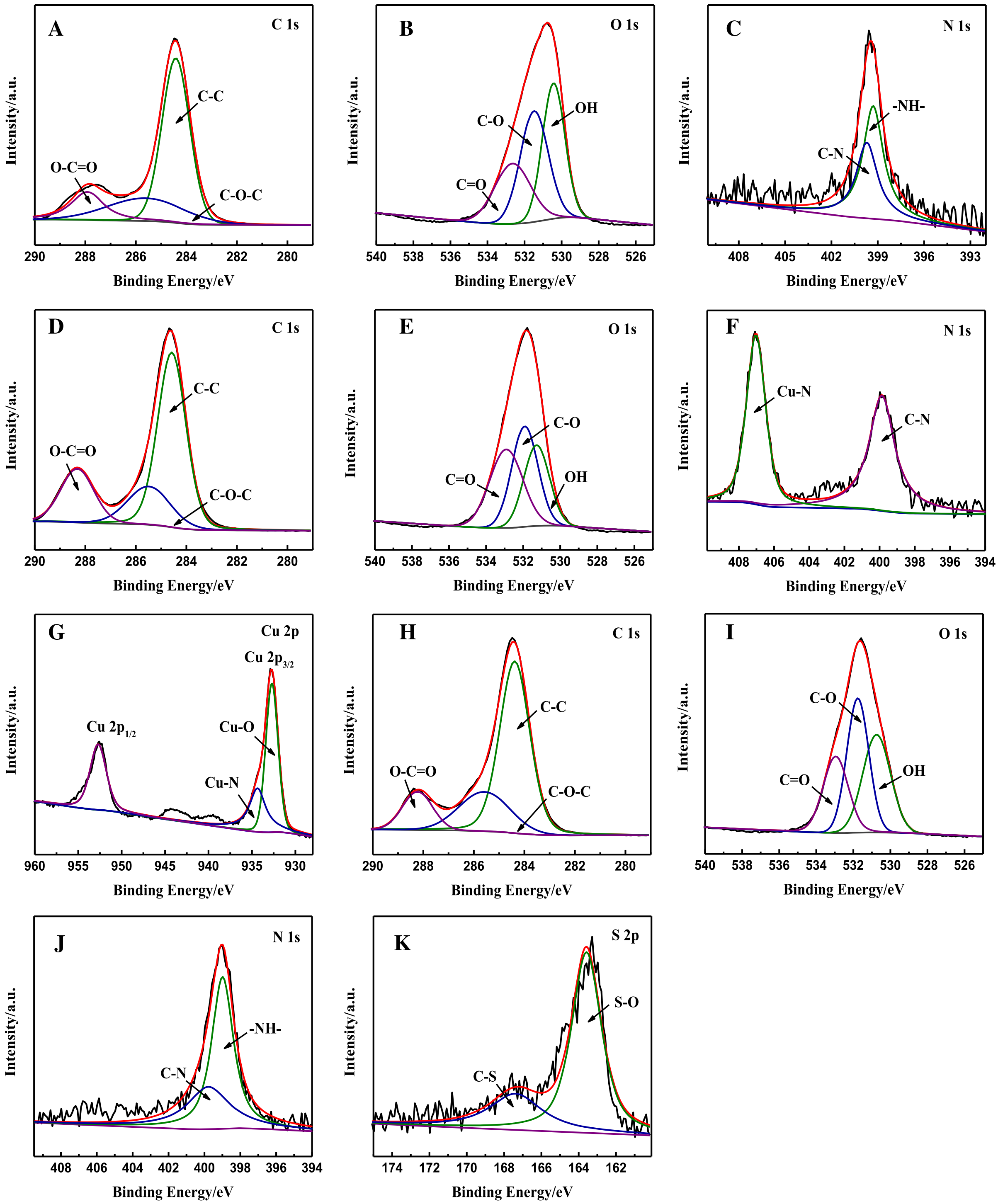


Fig. 5 C 1 s (**a**), O 1 s (**b**), and N 1 s (**c**) XPS spectra of the PAA/CMC/AC; C 1 s (**d**), O 1 s (**e**), N 1 s (**f**) and Cu 2p (**g**) XPS spectra of the PAA/CMC/AC-Cu^{2+}; C 1 s (**h**), O 1 s (**i**), N 1 s (**j**) and S 2p (**k**) XPS spectra of the PAA/CMC/AC-MB

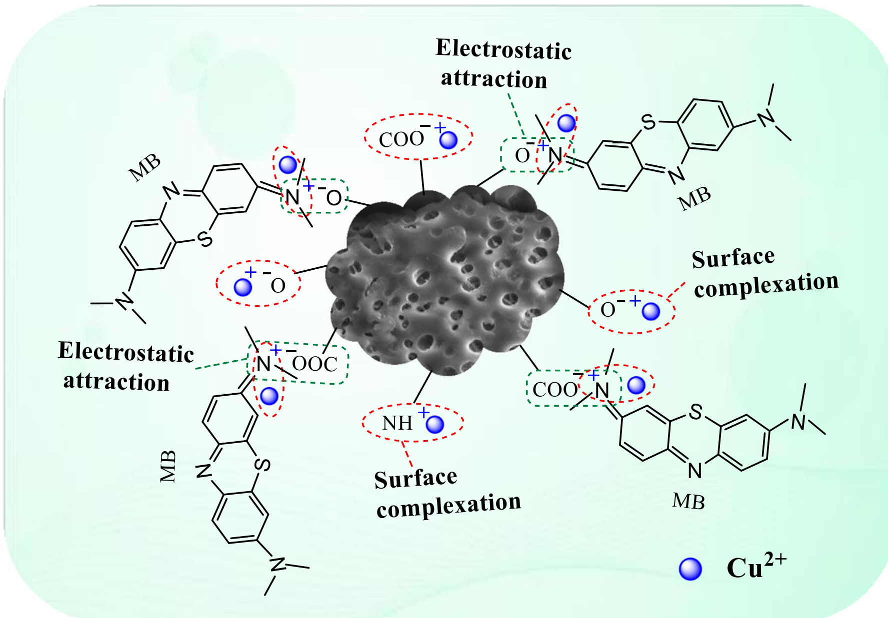


Fig. 6 The proposed adsorption mechanism for Cu^{2+} and MB

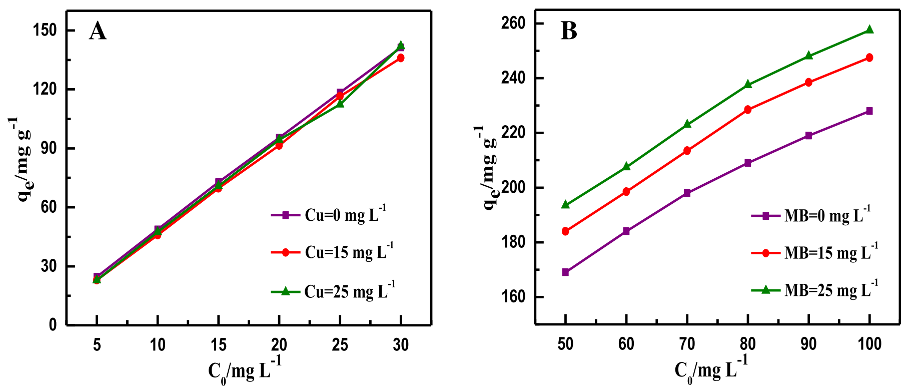


Fig. 7 Adsorption capacity of MB by PAA/CMC/AC in Cu^{2+}-MB system (**a**); adsorption capacity of Cu^{2+} by PAA/CMC/AC in MB-Cu^{2+} system (**b**)

Recycling performance of adsorbent

An ideal adsorbent can not only have excellent adsorption performance but also have stability and reusability performance. Considering the actual application situation, the adsorbent is subjected to multiple adsorption–desorption to determine its reusability, the results can be obtained from Fig. 8. The adsorption efficiency of PAA/CMC/AC for Cu^{2+} and MB can still reach 80% and 76% of the initial adsorption capacity, mainly because the Cu^{2+} and MB that have not been desorbed still occupy a certain number of adsorption sites. Simply put, PAA/CMC/AC as a highly efficient adsorbent that can be reused can achieve the purification of polluted water while also effectively recycle pollutants (Fan et al. 2020; Ngambia et al. 2019).

Conclusion

PAA/CMC/AC adsorbent was successfully prepared by grafting carboxymethyl cellulose and activated carbon onto polyacrylic acid chain to achieve the

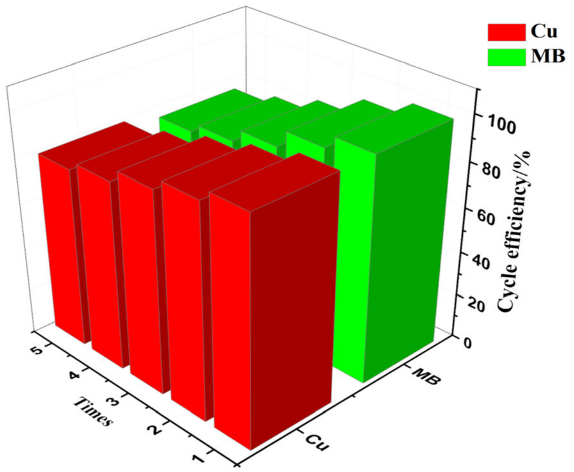


Fig. 8 Recycling performance of /PAACMC/AC during five cycles

removal of pollutants in independent system and selective removal in multiple system. As a result, the adsorption capacity for Cu^{2+} increased from 169 mg g^{-1} in independent system to 193.5 mg g^{-1} in multiple system, while the adsorption capacity for MB remained balanced, and the adsorption capacity was always 119.9 mg g^{-1}. All the results can prove that PAA/CMC/AC has a good adsorption capacity for these two pollutants, and pseudo-second-order kinetics and Freundlich isotherm can fit the adsorption data well. Additionally, the adsorbent has good recycling function and can be used as a promising adsorbent for practical application. In short, the PAA/CMC/AC adsorbent prepared by this research can provide some ideas and insights for the study of adsorbents for the treatment of wastewater containing heavy metal ions and dyes.

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Declarations

Conflict of interest The authors all declare there is no conflict over interest.

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